

Subject: IND320

Student name: Abdul Haadi Sheikh

Date: 28/09/2025

In this first part of the project I worked with both Jupyter Notebook and Streamlit to get a basic dashboard up and running. In Jupyter Notebook I read the supplied CSV file with Pandas, printed the contents in a structured way and created plots of both individual columns and all columns together. I made sure to run all code cells so the results and figures are visible when I export the notebook to PDF. I also added comments so the code is easier to understand and follow if someone else were to continue my work.

For the Streamlit app I created four pages, with a home page, a data table page, a plot page and an about page. The home page has a sidebar that lets you navigate to the other pages. On the data table page the CSV file is shown together with a small line chart for each column for the first month of the dataset. On the plot page it is possible to choose one column or all columns from a drop down menu and also filter months with a slider. The app uses caching so that the CSV file loads faster, and I added the CSV file into the repository so that it works both locally and on Streamlit Cloud.

I struggled a lot with file formats in the beginning. I first worked only with ipynb files but Streamlit does not accept those, so I had to make sure to keep the code in py files as well. I also had problems when I tried to organize everything into folders, because Streamlit Cloud and GitHub did not always find the files in the right place. In the end I fixed this by keeping streamlit_app.py and utils.py in the root folder, and by using relative paths instead of absolute ones for loading the CSV file. I also had to clean up the requirements.txt file so that only the packages I really needed were listed. After these changes the deployment on Streamlit Cloud finally worked.

I used AI tools like ChatGPT while working on the project. It helped me when I got stuck on errors, for example with missing files, wrong imports and when the app failed to run in the cloud. It also explained how to set up caching, how to structure the repository and how to read the Streamlit error logs. I learned more about how to debug and organize a project like this.

My GitHub repository is available here: https://github.com/Sheikh2003/IND320_1_Project

My Streamlit app is available here: <https://streamspinn.streamlit.app>

streamlit_app

September 28, 2025

```
[24]: import streamlit as st # Imports the Streamlit library
      from utils import load_data # Imports the function to load data from a utility_
      ↪module.
```

```
[25]: st.set_page_config(page_title="Streamlit MWE", page_icon=" ", layout="wide") #_
      ↪Sets the overall configuration
```

2025-09-28 15:09:26.431 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

```
[26]: st.title("Open-Meteo: Overview") # Displays the main title of the application.
      st.caption("Explore the `open-meteo-subset.csv` dataset. Use the menu on the_
      ↪left to view the data table with sparklines or an interactive plot.")
```

2025-09-28 15:09:26.439 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:09:26.440 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:09:26.440 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:09:26.441 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:09:26.442 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:09:26.442 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

```
[26]: DeltaGenerator()
```

```
[27]: # Sidebar navigation
      st.sidebar.header("Navigation") # Creates a header in the sidebar for the_
      ↪navigation links.
      st.sidebar.page_link("pages/2_Data_Table.py", label=" Data table") # Adds a_
      ↪link to the "Data table" page
      st.sidebar.page_link("pages/3_Plot.py", label=" Plot") # Adds a link to the_
      ↪"Plot" page
      st.sidebar.page_link("pages/4_About.py", label=" About") # Adds a link to the_
      ↪"About" page
```

```

2025-09-28 15:09:26.451 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:09:26.452 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:09:26.453 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:09:26.454 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:09:26.454 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:09:26.455 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:09:26.455 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:09:26.456 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:09:26.457 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:09:26.457 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:09:26.457 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:09:26.458 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:09:26.459 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:09:26.459 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:09:26.459 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.

```

```
[27]: DeltaGenerator(_root_container=1, _parent=DeltaGenerator())
```

```

[28]: def load_data():
    # Sjekk først i samme mappe som utils.py
    path_local = os.path.join(os.path.dirname(__file__), "open-meteo-subset.
↪csv")
    if os.path.exists(path_local):
        path = path_local
    else:
        # Hvis ikke, sjekk i prosjektroten (en mappe opp)
        path_root = os.path.join(os.path.dirname(__file__), "..",
↪"open-meteo-subset.csv")
        if os.path.exists(path_root):
            path = path_root
        else:

```

```

        raise FileNotFoundError("Could not find 'open-meteo-subset.csv' in_
↳either current folder or project root")

    df = load_data() # Loads the data into a DataFrame
    st.subheader("Quick look at the data") # Displays a subheading for the data_
↳preview.
    st.dataframe(df.head(), use_container_width=True) # Shows the first five_
↳rows of the DataFrame
    st.success(f"Loaded {len(df)} rows × {len(df.columns)} columns.") #_
↳Displays a success message

```

utils

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```
[1]: import os
import pandas as pd
import streamlit as st

    # Cache the function's output to improve performance in Streamlit
    ↪applications.
@st.cache_data(show_spinner=False)
def load_data():
    # Construct the full file path dynamically.
    # Read the CSV file into a DataFrame.
    path = os.path.join(os.path.dirname(__file__), "open-meteo-subset.csv")
    df = pd.read_csv(path)

    # Attempt to find a date/time column in the DataFrame.
    date_like = [c for c in df.columns if any(k in c.lower() for k in ["date",
    ↪"time", "datetime", "timestamp"])]
    if date_like:
        # If a potential date/time column is found, select the first one.
        dcol = date_like[0]
        # Convert the selected column to datetime objects.
        df[dcol] = pd.to_datetime(df[dcol], errors="coerce")
        # Check if any successful datetime conversion occurred.
        if df[dcol].notna().any():
            df = df.sort_values(dcol)
            # Create a new 'month' column by converting the datetime to a monthly
            ↪period string
            df["month"] = df[dcol].dt.to_period("M").astype(str)
        else:
            df["month"] = "Unknown"

    return df
```

2025-09-28 15:03:24.538 WARNING streamlit.runtime.caching.cache_data_api: No runtime found, using MemoryCacheStorageManager

2_Data_Table

September 28, 2025

```
[1]: import streamlit as st
import pandas as pd
from utils import load_data
```

2025-09-28 15:24:36.074 WARNING streamlit.runtime.caching.cache_data_api: No runtime found, using MemoryCacheStorageManager

```
[2]: st.set_page_config(page_title="Data Table", page_icon=" ", layout="wide") # Sets the page title and uses a wide layout.
st.title("Data Table") # Displays the main header.
```

2025-09-28 15:24:36.115 WARNING
streamlit.runtime.scriptrunner_utils.script_run_context: Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2025-09-28 15:24:36.116 WARNING
streamlit.runtime.scriptrunner_utils.script_run_context: Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2025-09-28 15:24:36.253

Warning: to view this Streamlit app on a browser, run it with the following command:

```
streamlit run /Users/a.h.sheikh/.pyenv/versions/ind320/lib/python3.12/site-packages/ipykernel_launcher.py [ARGUMENTS]
```

2025-09-28 15:24:36.253 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2025-09-28 15:24:36.253 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

```
[2]: DeltaGenerator()
```

```
[3]: df = load_data() # Loads the DataFrame
```

2025-09-28 15:24:36.261 No runtime found, using MemoryCacheStorageManager

```
[4]: st.subheader("Raw data") # Displays a subheading.
st.dataframe(df, use_container_width=True) # Displays the full raw DataFrame in an interactive Streamlit table.
```

2025-09-28 15:24:36.331 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
 2025-09-28 15:24:36.333 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
 2025-09-28 15:24:36.333 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
 2025-09-28 15:24:36.334 Please replace `use\_container\_width` with `width`.

`use\_container\_width` will be removed after 2025-12-31.

For `use\_container\_width=True`, use `width='stretch'`. For
 `use\_container\_width=False`, use `width='content'`.
 2025-09-28 15:24:36.350 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
 2025-09-28 15:24:36.352 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
 2025-09-28 15:24:36.352 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

[4]: DeltaGenerator()

```
[5]: # Find datetime-like and numeric columns
date_cols = [c for c in df.columns if any(k in c.lower() for k in ["date",
↳ "time", "datetime", "timestamp"])] # Identifies columns suitable for time
↳ series analysis.
date_col = date_cols[0] if date_cols else None
num_cols = [c for c in df.columns if pd.api.types.is_numeric_dtype(df[c])]
```

```
[6]: # Show row-wise sparklines
if "month" in df.columns and date_col and num_cols: # Checks if month, date,
↳ numeric data are met.
    months = sorted([m for m in df["month"].dropna().unique()])
    if months:
        first_month = months[0]
        mdf = df[df["month"] == first_month].copy() # Filters the data to
↳ include only the first month.

        # Prepare data for the sparkline table: each numeric column becomes a
↳ row.
        rows = [{"Metric": col, "First month trend": mdf[col].tolist()} for col
↳ in num_cols]
        table = pd.DataFrame(rows)

        # Calculate global min/max across all numeric columns in the first
↳ month.
        y_min = float(pd.concat([mdf[c] for c in num_cols]).min(skipna=True))
        y_max = float(pd.concat([mdf[c] for c in num_cols]).max(skipna=True))
```

```

        st.subheader(f"First month - sparklines ({first_month})") # Displays a
        ↪ subheading for the sparkline table.
        st.dataframe(
            table,
            column_config={
                "Metric": st.column_config.TextColumn("Metric"), # Configures
                ↪ the 'metric' column as standard text.
                "First month trend": st.column_config.LineChartColumn(
                    "First month trend",
                    y_min=y_min, y_max=y_max, # Ensures all sparklines share
                    ↪ the same y-axis scale for comparison.
                    help="Row-wise sparkline for the first month",
                ),
            },
            use_container_width=True,
            hide_index=True,
        )
    else:
        st.info("No month found to display sparklines.") # Fallback if month
        ↪ column exists but is empty.
else:
    st.info("A date/time column and numeric columns are required to build
    ↪ sparklines.") # Fallback if necessary data types are missing.

```

2025-09-28 15:24:36.370 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:24:36.372 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:24:36.372 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:24:36.372 Please replace `use\_container\_width` with `width`.

`use\_container\_width` will be removed after 2025-12-31.

For `use\_container\_width=True`, use `width='stretch'`. For

`use\_container\_width=False`, use `width='content'`.

2025-09-28 15:24:36.374 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:24:36.374 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:24:36.374 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:24:36.372 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:24:36.372 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:24:36.372 Please replace `use\_container\_width` with `width`.

`use\_container\_width` will be removed after 2025-12-31.

For `use\_container\_width=True`, use `width='stretch'`. For

`use\_container\_width=False`, use `width='content'`.

2025-09-28 15:24:36.374 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:24:36.374 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:24:36.374 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

3_Plot

September 28, 2025

```
[4]: import streamlit as st
import pandas as pd
import plotly.express as px
from utils import load_data
```

```
[5]: st.set_page_config(page_title="Plot", page_icon=" ", layout="wide") # Sets the
    ↪page configuration to use a wide layout for the chart.
st.title("Data plot (Plotly)") # Displays the main header
```

2025-09-28 15:24:28.266 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:24:28.268 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:24:28.269 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:24:28.269 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

```
[5]: DeltaGenerator()
```

```
[6]: # Load data (cached in utils.load_data)
df = load_data() # Loads the full dataset,
```

```
[7]: # Find a date/time-like column
date_cols = [c for c in df.columns if any(k in c.lower() for k in ["date",
    ↪"time", "datetime", "timestamp"])] # Finds column names that indicate time
    ↪series data.
date_col = date_cols[0] if date_cols else None
if date_col is None: # Checks if a primary time column was successfully
    ↪identified.
    st.error("No date/time column found - cannot build a time series.") #
    ↪Displays an error if the X-axis data is missing.
    st.stop() # Halts execution since the core plot cannot be generated.
```

```
[8]: # Ensure datetime dtype
if not pd.api.types.is_datetime64_any_dtype(df[date_col]):
    df[date_col] = pd.to_datetime(df[date_col], errors="coerce") # Converts the
    ↪time column to the required datetime object type.
```

```
[9]: # Numeric columns
num_cols = [c for c in df.columns if pd.api.types.is_numeric_dtype(df[c])] #
↳ Filters the columns to find all available numeric data for plotting.
if not num_cols:
    st.error("No numeric columns found to plot.") # Displays an error if there
↳ are no Y-axis.
    st.stop() # Stops execution.
```

```
[10]: # Select columns to plot
choice = st.selectbox("Select column(s) to plot", ["All columns"] + num_cols,
↳ index=0) # Creates an interactive dropdown menu for column selection.
```

```
2025-09-28 15:24:28.378 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.387 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.387 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.388 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.389 Session state does not function when running a script
without `streamlit run`
2025-09-28 15:24:28.389 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.389 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.390 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.387 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.387 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.388 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.389 Session state does not function when running a script
without `streamlit run`
2025-09-28 15:24:28.389 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.389 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.390 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
```

```
[11]: # Select month range
if "month" in df.columns and df["month"].notna().any(): # Checks for the
↳ pre-calculated 'month' column
    months = sorted(df["month"].dropna().unique().tolist())
```

```

    default_month = months[0]
    start, end = st.select_slider("Select month(s)", options=months,
    ↪value=(default_month, default_month)) # Creates an interactive slider
    mask = (df["month"] >= start) & (df["month"] <= end)
    pdf = df.loc[mask].copy()
    subtitle = f"Months: {start} to {end}"
else:
    pdf = df.copy() # Uses the entire dataset if no 'month' column exists.
    subtitle = "All rows"

```

```

2025-09-28 15:24:28.398 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.399 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.399 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.399 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.400 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.400 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.399 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.399 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.399 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.400 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:24:28.400 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.

```

```

[12]: # Build Plotly figure
if choice == "All columns": # Executes the logic for plotting multiple series.
    long_df = pdf[[date_col] + num_cols].melt(id_vars=date_col,
    ↪var_name="column", value_name="value") # Pivots the table from wide to long
    ↪format, which Plotly requires for multi-line plots.
    fig = px.line(long_df, x=date_col, y="value", color="column", # Creates the
    ↪Plotly line chart with multiple colored lines.
                  title=f"Time series - {subtitle}",
                  labels={date_col: "Date", "value": "Value", "column":
    ↪"Column"})
else: # Executes the logic for plotting a single series.
    fig = px.line(pdf, x=date_col, y=choice, # Creates a standard Plotly line
    ↪chart for the selected column.
                  title=f"Time series - {choice} - {subtitle}",

```

```
labels={date_col: "Date", choice: "Value"})
```

```
[13]: st.plotly_chart(fig, use_container_width=True) # Renders the Plotly figure
      ↪ within the Streamlit app, adjusting to the container width. allowing for
      ↪ fast reloading on subsequent accesses.
```

```
2025-09-28 15:24:28.495 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
```

```
2025-09-28 15:24:28.496 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
```

```
2025-09-28 15:24:28.500 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
```

```
2025-09-28 15:24:28.502 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
```

```
2025-09-28 15:24:28.503 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
```

```
[13]: DeltaGenerator()
```

4__About

September 28, 2025

```
[1]: import streamlit as st # Imports the Streamlit library.
```

```
[2]: st.set_page_config(page_title="About", page_icon=" ") # Configures the browser
      ↪ tab title
      st.title("About this app") # Displays the main title
```

```
2025-09-28 15:09:37.451 WARNING
streamlit.runtime.scriptrunner_utils.script_run_context: Thread 'MainThread':
missing ScriptRunContext! This warning can be ignored when running in bare mode.
2025-09-28 15:09:37.452 WARNING
streamlit.runtime.scriptrunner_utils.script_run_context: Thread 'MainThread':
missing ScriptRunContext! This warning can be ignored when running in bare mode.
2025-09-28 15:09:37.573
Warning: to view this Streamlit app on a browser, run it with the
following
command:
```

```
streamlit run /Users/a.h.sheikh/.pyenv/versions/ind320/lib/python3.12/site-
packages/ipykernel_launcher.py [ARGUMENTS]
2025-09-28 15:09:37.573 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2025-09-28 15:09:37.573 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
```

```
[2]: DeltaGenerator()
```

```
[3]: st.markdown("""
      **In short:** A small, multi-page Streamlit app for exploring the
      ↪ `open-meteo-subset.csv` dataset.

      **What you can do**
      - Browse the raw data table.
      - See per-row sparklines for the first month (using `st.column_config.
        ↪ LineChartColumn`).
      - Build interactive time-series plots with **Plotly** - select a single column
        ↪ or all columns, and filter by month range.

      **How it works**
```

```
- Reads a **local CSV** and caches it with `st.cache_data` for speed.
- Pages: Home, Data, Plot, About.
- Dependencies (see `requirements.txt`): `streamlit`, `pandas`, `numpy`,  
  ↳ `plotly`.
"""
```

2025-09-28 15:09:37.579 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:09:37.580 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

2025-09-28 15:09:37.581 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

[3]: DeltaGenerator()